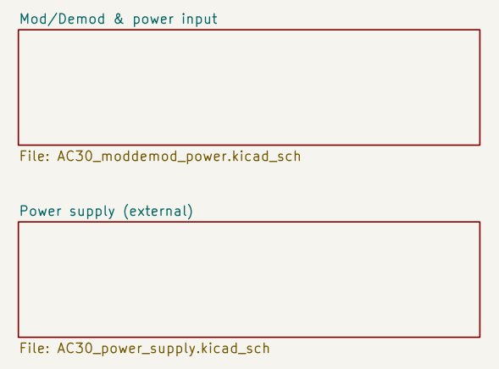
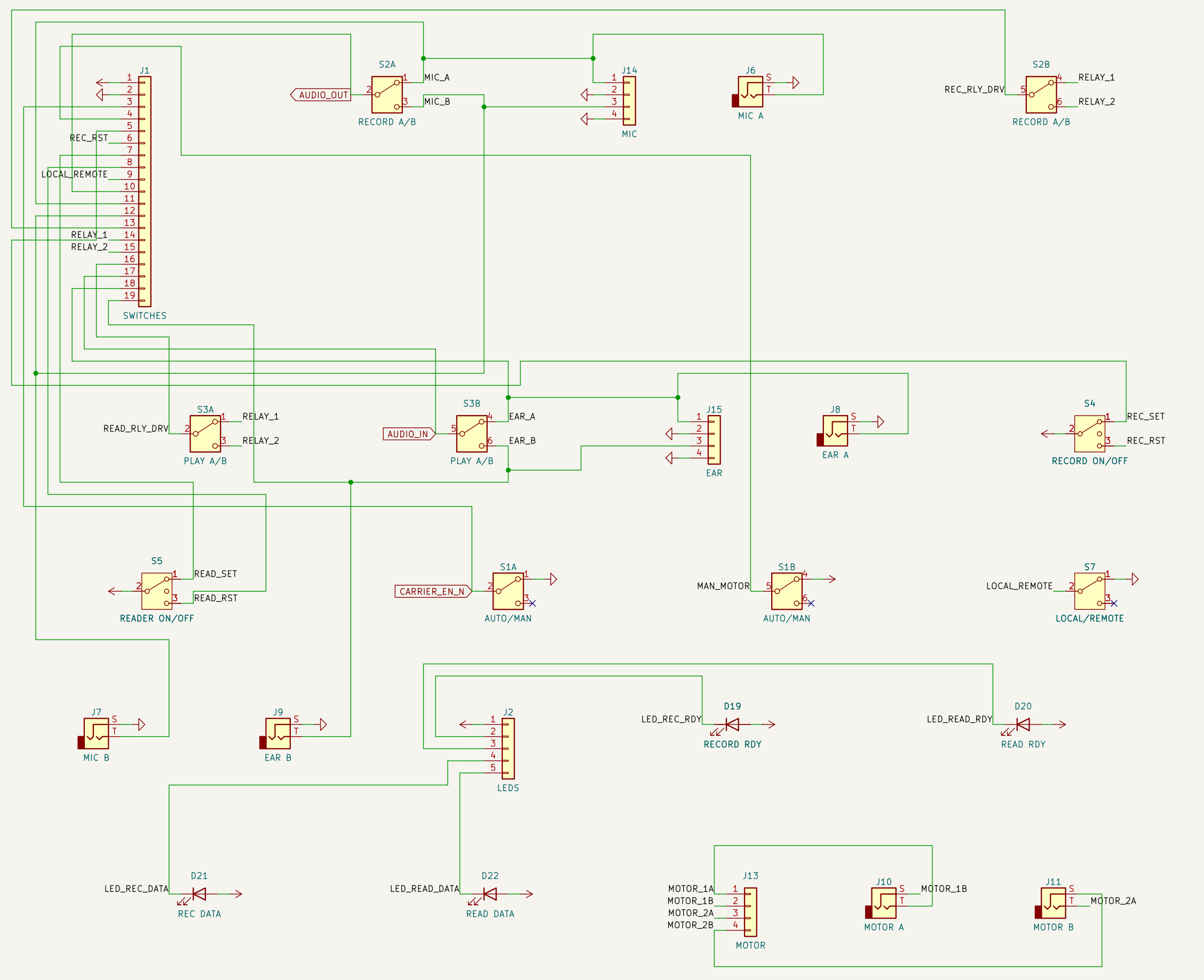
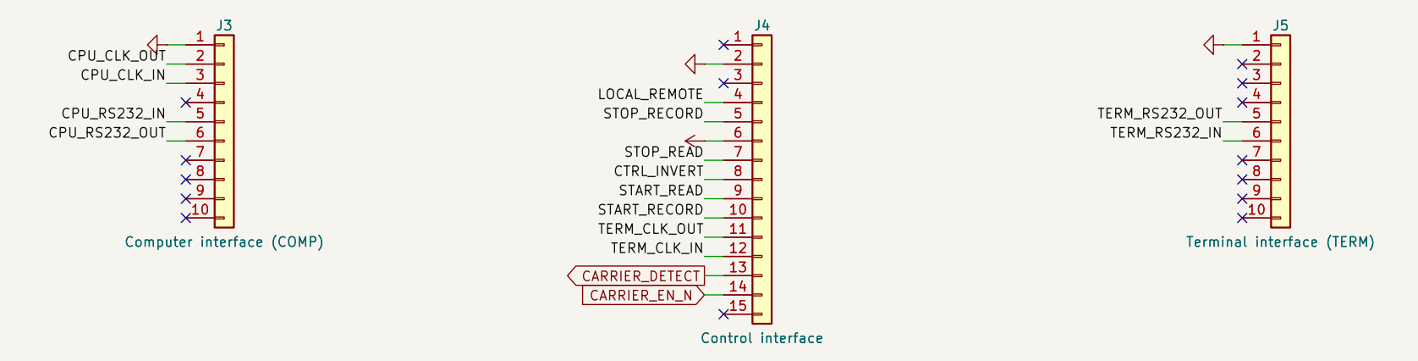
Generated from docs/ac30_schematica.jpg (SWTPC AC-30 Cassette Tape Switching Schematic),
docs/AC30-BOM.md and docs/ac30_wiring1a.jpg. Each block is wired; net labels join the blocks and sheets
(identical labels are connected).
Transcription notes / assumptions:
- Drawing pin-number slips corrected to the real pinouts: IC7D drawn 8->9 (4049: 9->10);
  IC13C drawn inputs 7,8 (4070: 8,9).
- 4053 INH (pin 6) and VEE (pin 7) are not shown on the drawing; tied to GND.
- IC10 pins 4/8 tie to the R38 supply node, assumed +5V.
- IC15 (1488) supply pins are not shown; taken as +13V / -13V from the power supply (sheet 3).
- CONTROL INVERT: drawn as R42 pull-up; it is also brought out on J4.
- J3-J5 pin positions match the original artwork (docs/ac30_redblue150.jpg).
- The front panel is wired to five rows of 0.1 in through-hole pads instead of the artwork's
  J1/J2 harness headers: J1 SWITCHES, J2 LEDS, J13 MOTOR, J14 MIC (MIC A/B jacks) and
  J15 EAR (EAR A/B jacks). The S2/S3 audio poles wire to the SWITCHES pads (AUDIO_OUT,
  MIC_A/B, AUDIO_IN, EAR_A/B); the board joins them.
- MIC/EAR jack reference designators J6-J9 are assigned arbitrarily; J10/J11 are the motor jacks.
- Trimmer R39 (DELAY, next to IC10 on the artwork) is not on any schematic; it is listed,
  not placed, in tools/netlist.py.
- The artwork's J2 pins 11/12 (18 VAC A/B) are gone: the power supply is external.


Mod/Demod & power input
File: AC30_moddemod_power.kicad_sch
Power supply (external)
File: AC30_power_supply.kicad_sch

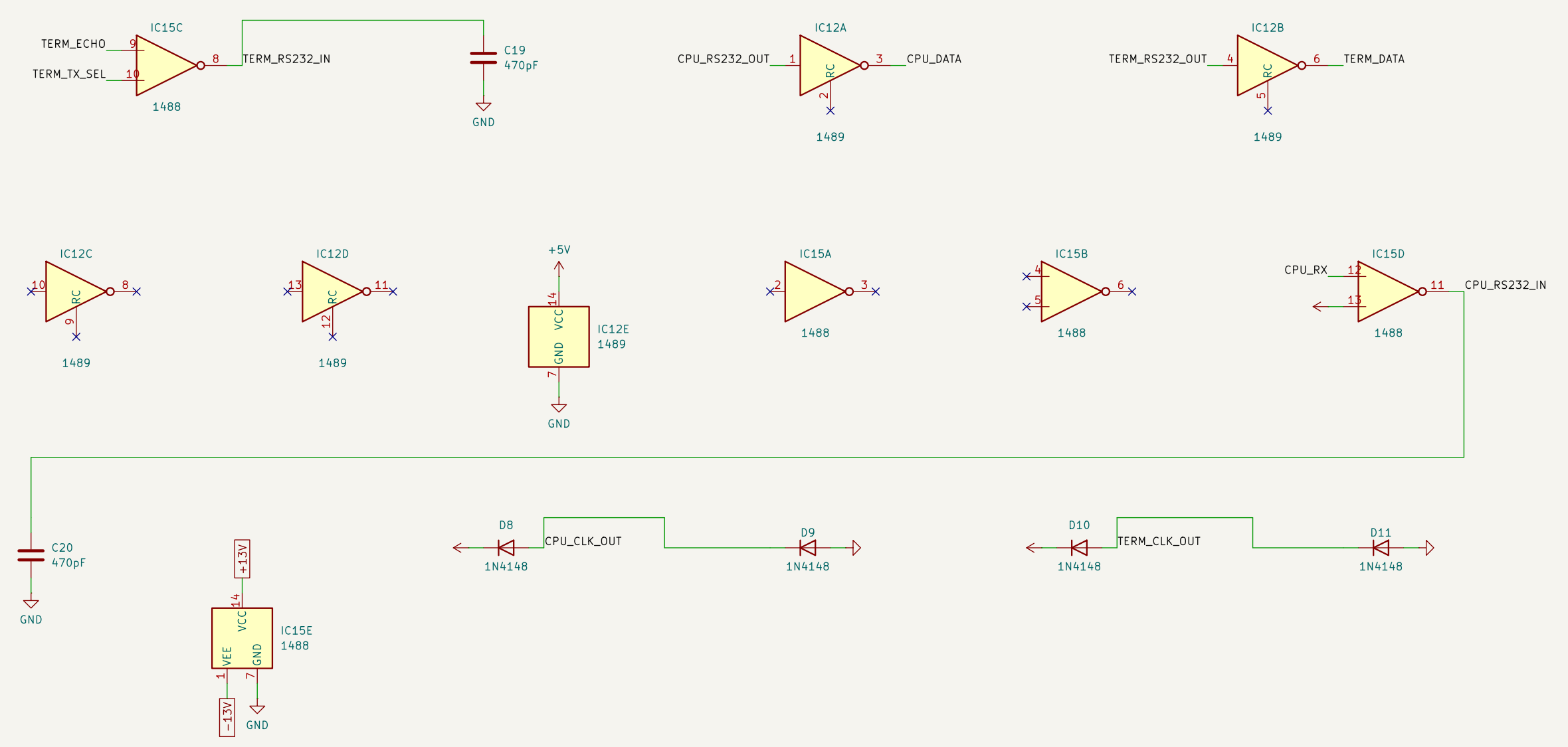
Front panel switches, LEDs and jacks



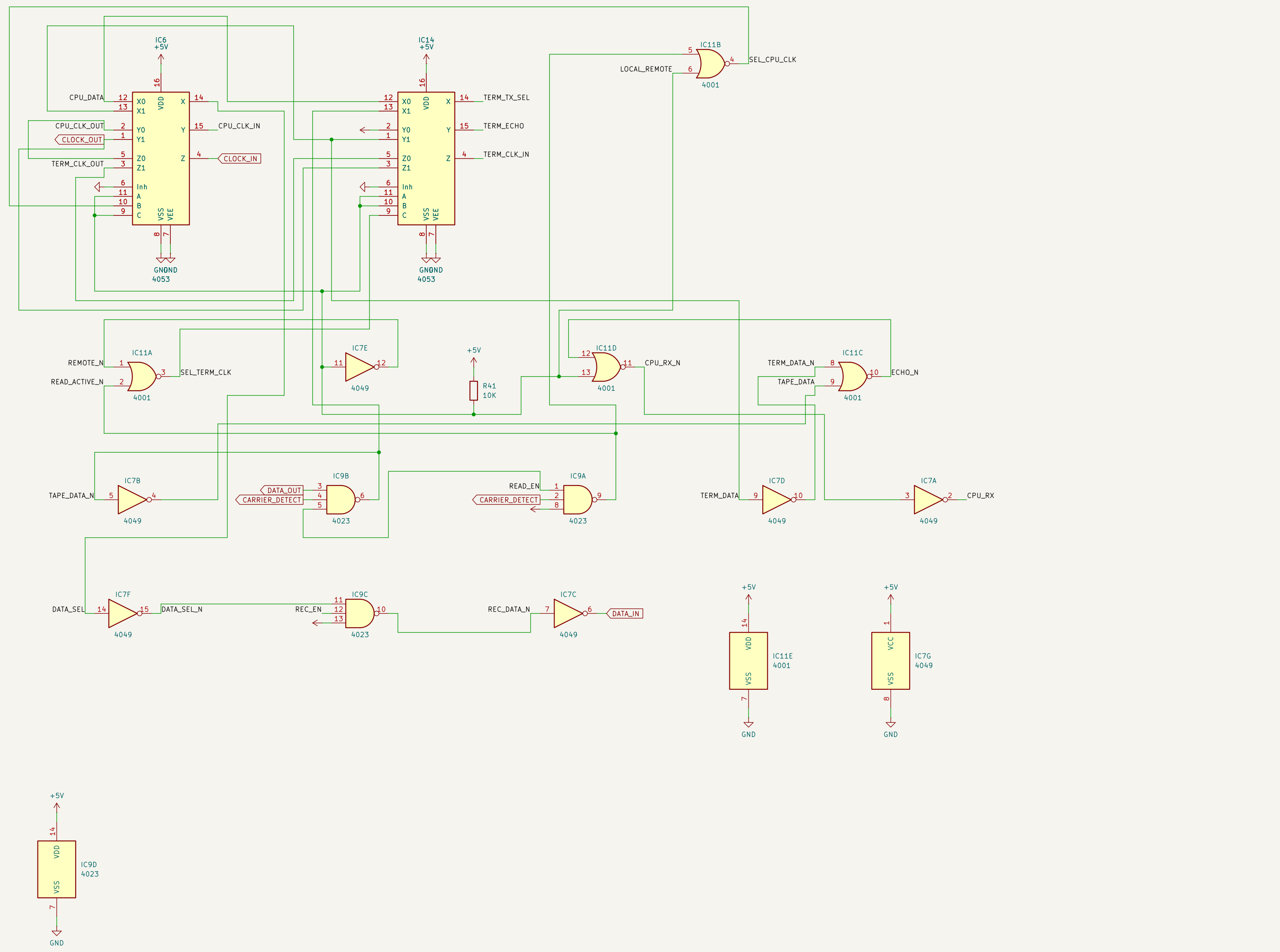
Board connectors (pin positions per original artwork)



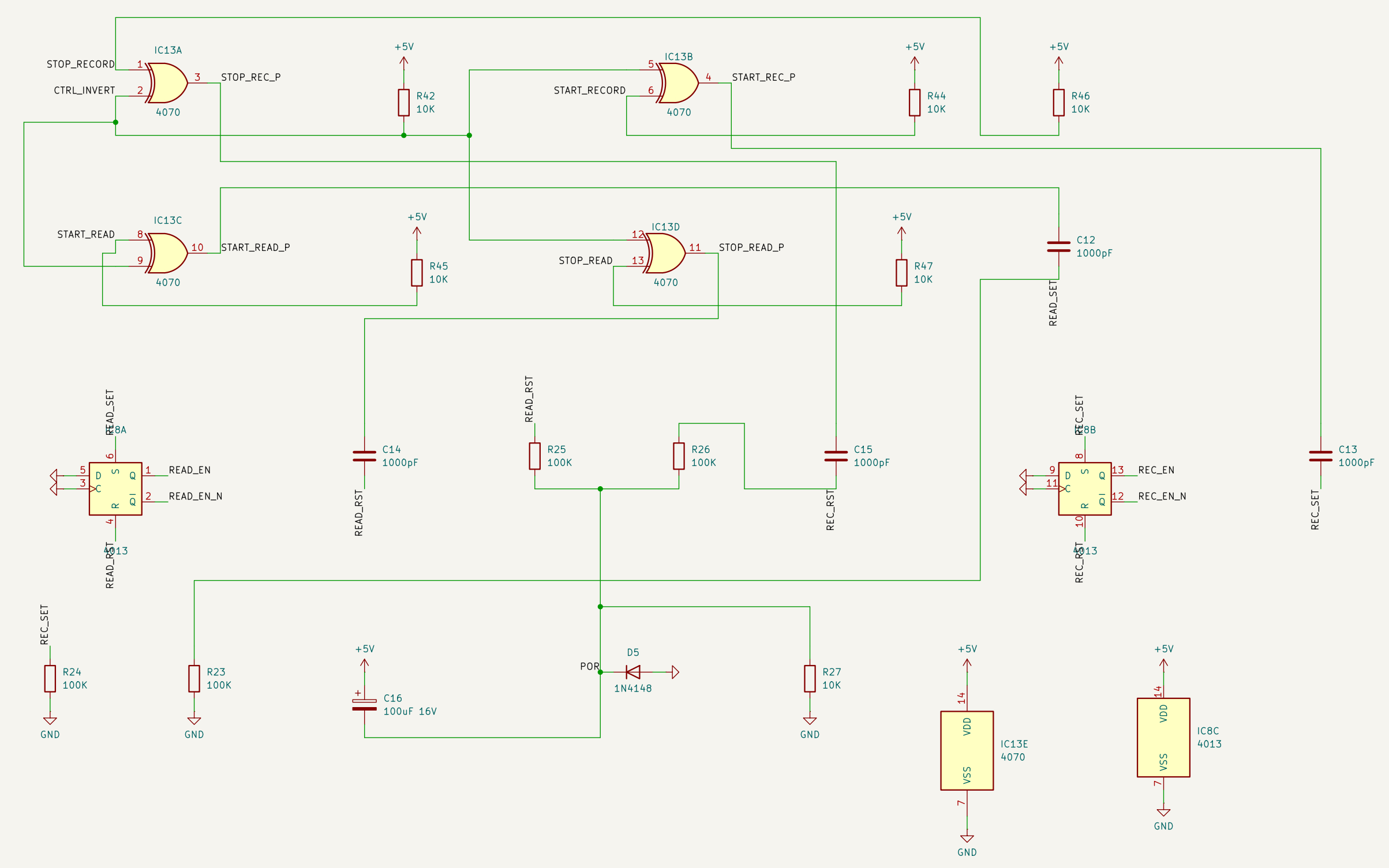
RS-232 receivers / drivers, clock clamps



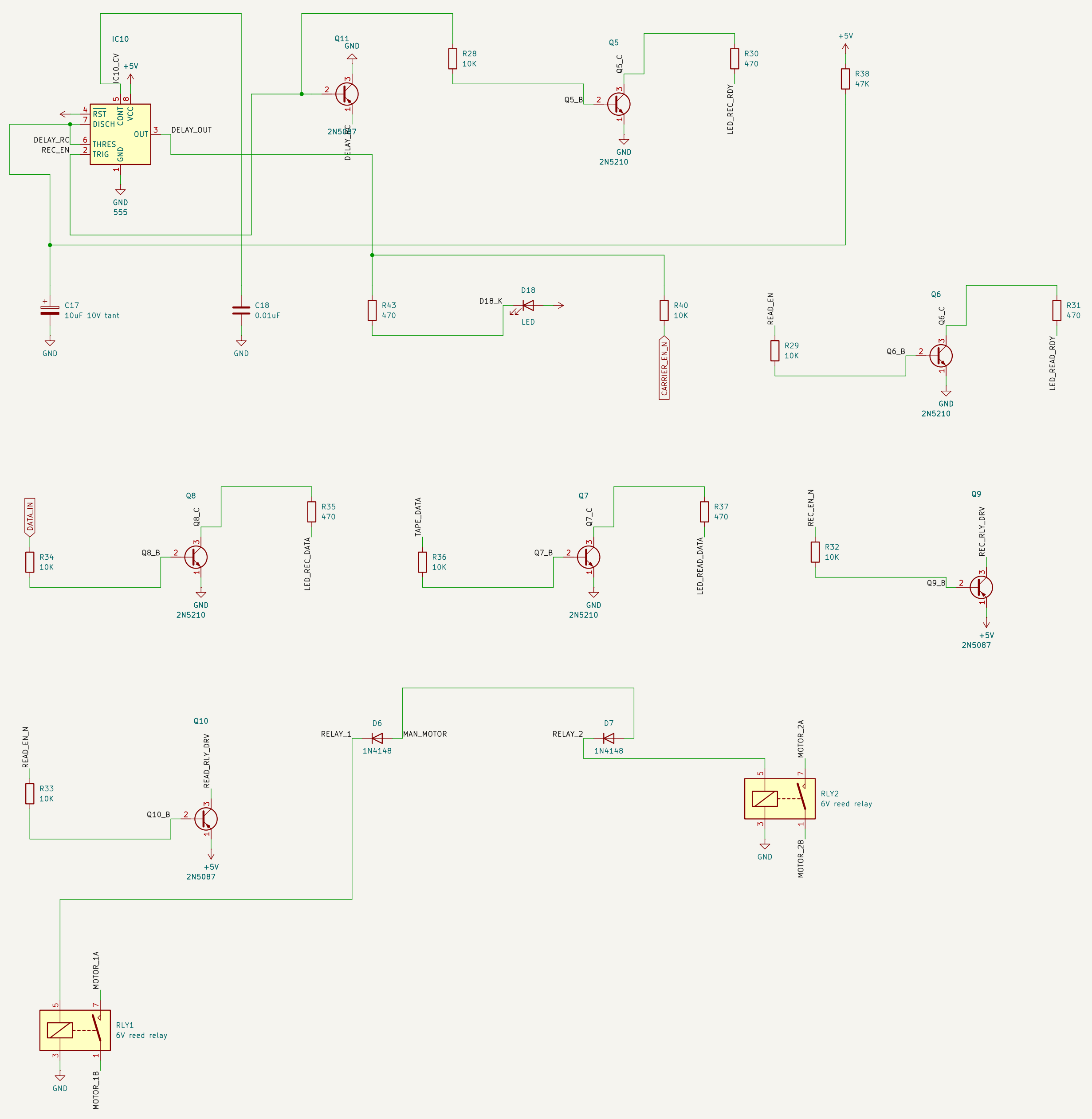
Local/Remote steering and data/clock selectors



Record / read control latches and power-on reset



LED drivers, relay drivers, carrier-enable delay

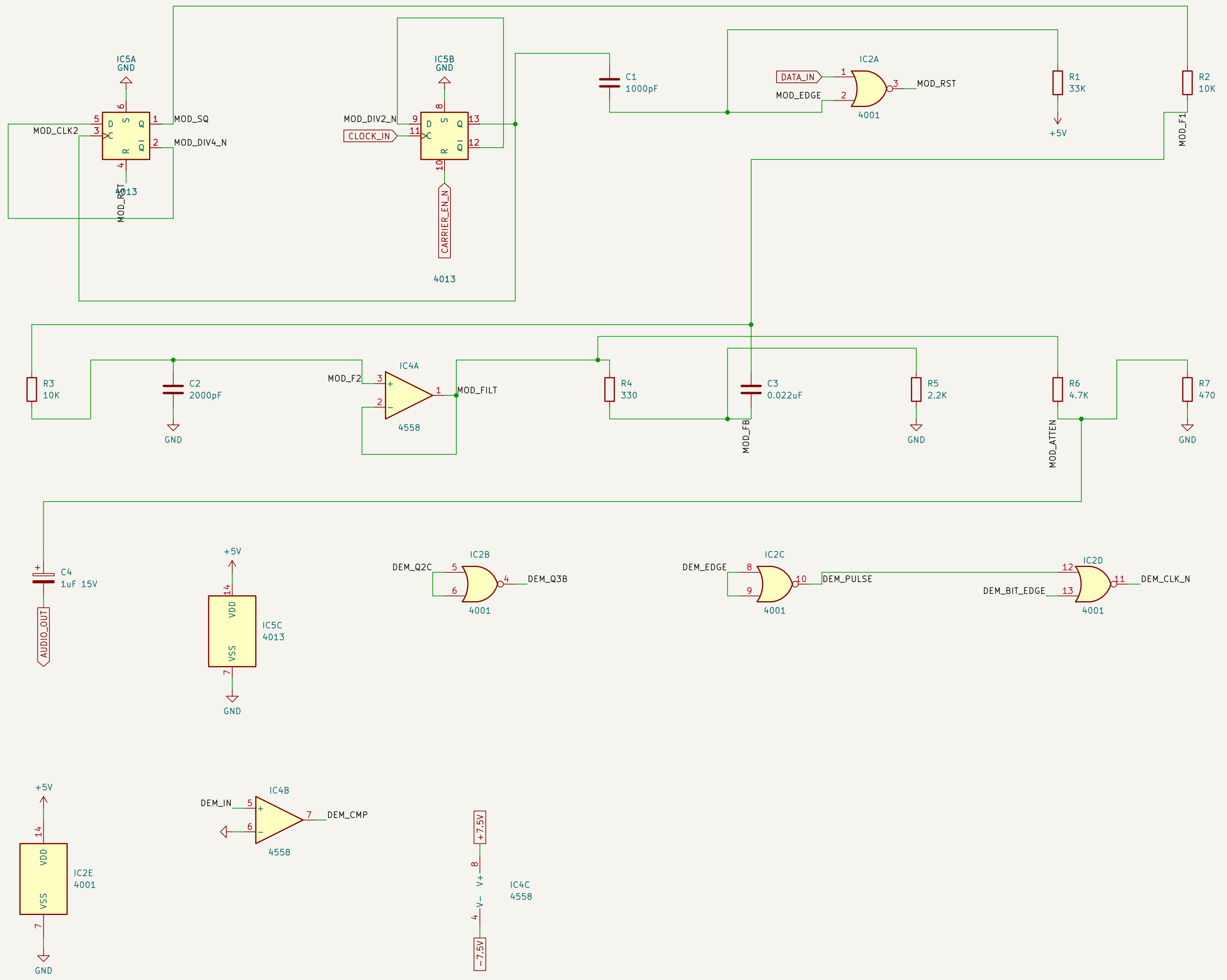


Generated by kicad/tools/gen.py — edit netlist.py and regenerate, or edit in KiCad
Source: docs/ac30_schematica.jpg, docs/ac30_mod_demod.pdf, docs/ac30_PS.pdf, docs/AC30-BOM.m
SWTPC AC-30 reproduction
Sheet: /
File: AC30.kicad_sch
Title: SWTPC AC-30 Cassette Interface - Switching
Size: A0 Date: 2026-10-08 **Rev: A**
KiCad E.D.A. 10.0.7 Id: 1/3

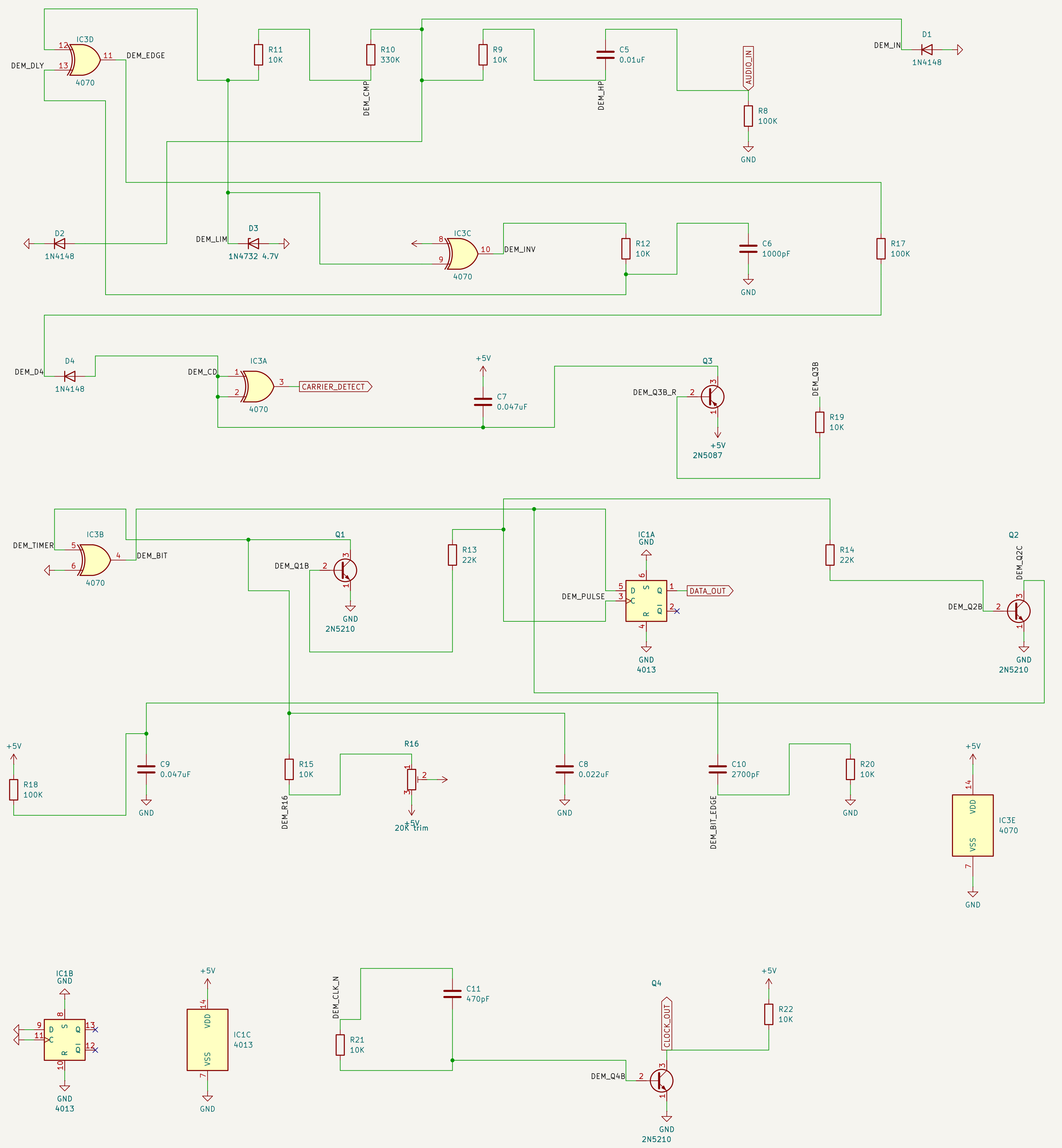
Generated from docs/ac30_mod_demod.pdf (SWTPC AC–30 Cassette Tape Modulator/Demodulator
Schematic). Each block is wired; net labels join the blocks and sheets (identical labels are connected).
Transcription notes / assumptions:
- IC4 (4558) supply pins are not drawn. The assembly manual says IC4 runs from the +/–7.5 V zener rails (D12/D13 on the power supply sheet).
- R16's far end has a terminal dot with no label; taken as +5V (Q1/C8 timer pull–up).
- IC1B is unused and not drawn: its inputs are tied to GND, its outputs left open.
- The manual's text calls the pulse inverter IC2A; the schematic (followed here) uses IC2C.

Power input: the supply (sheet 3) is built as a separate unit. Its six DC outputs come onto
the board on J12, a 6–pin KK–396 header (1 GND, 2 +5V, 3 +13V, 4 –13V, 5 +7.5V, 6 –7.5V). C23–C25 are +5 V bypass capacitors on the board
(drawn at the regulator on docs/ac30_PS.pdf; next to IC1, IC7 and IC13 on the original artwork).

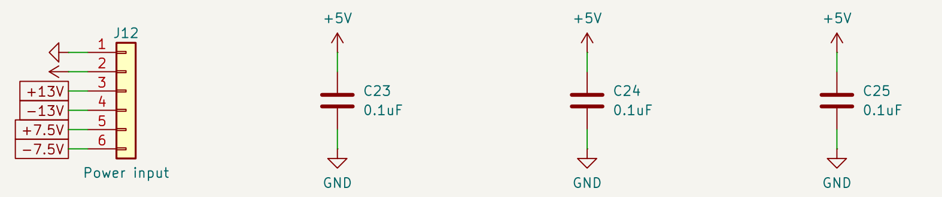
Modulator: CLOCK IN /2 (IC5B), /1 or /2 by DATA IN (IC5A), 2–pole filter IC4A



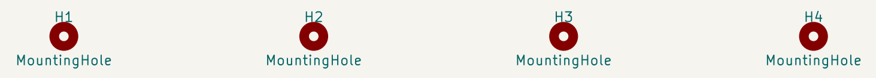
Demodulator: comparator IC4B, edge pulses IC3C/D, carrier detect, data and 16X clock out



Power input connector (from the external supply) and +5 V bypass

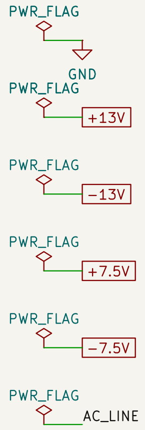


Mounting holes

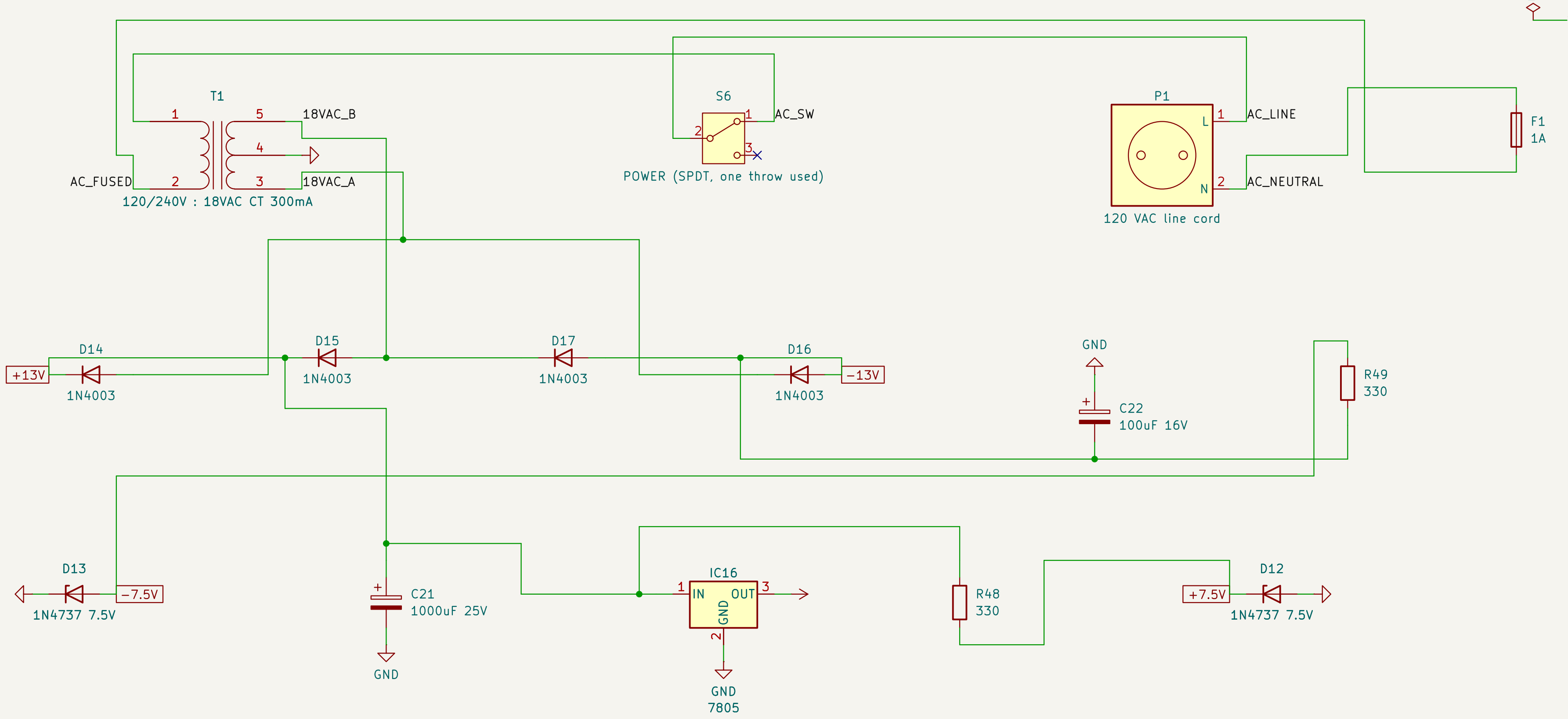


Source: docs/ac30_mod_demod.pdf; power input J12 for the external supply
SWTPC AC–30 reproduction
Sheet: /Mod/Demod & power input/
File: AC30_moddemod_power.kicad_sch
Title: SWTPC AC–30 – Modulator/Demodulator & power input
Size: A1
Date: 2026–10–08
Rev: A
KiCad E.D.A. 10.0.7
Id: 2/3

Generated from docs/ac30_PS.pdf (SWTPC AC-30 Audio Cassette Power Supply Schematic).
Built as an EXTERNAL unit: these parts have no footprints and are not on the PC board. Wire the six outputs to the board's power input J12 (KK-396: 1 GND, 2 +5V, 3 +13V, 4 -13V, 5 +7.5V, 6 -7.5V).
Transcription notes / corrections:
- The drawing joins the IC16 GND / C23–C25 return line to the bridge's negative corner (−13 V) through two crossings. That would put the 7805's ground at −13 V, so IC16 GND is taken as GND.
- T1's two 120 V primaries are drawn in parallel (120 VAC); drawn here as one primary. The 18 VAC secondary is centre-tapped, tap to GND: the bridge gives +13 V (C21) and −13 V (C22).
- D14–D17 numbering within the bridge is not shown on the drawing; assigned here.
- S6 is an SPDT toggle (BOM) used as an on/off switch; one throw is unused.
- IC15 (1488) supply pins are not drawn on the switching schematic; taken as +/−13 V, the only load for those outputs. IC4 (4558) uses +/−7.5 V (assembly manual).



Power supply (external): 120 VAC -> T1 -> bridge D14–D17 -> +/-13 V, zeners +/-7.5 V, 7805 +5 V



Source: docs/ac30_PS.pdf; parts have no footprints – built as a separate unit
SWTPC AC-30 reproduction
Sheet: /Power supply (external)/
File: AC30_power_supply.kicad_sch
Title: SWTPC AC-30 – Power supply (external, not on PCB)
Size: A3 Date: 2026-10-08 **Rev: A**
KiCad E.D.A. 10.0.7 Id: 3/3